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Intelligent Deep Learning Model for Securing KMS Through CSF Deviation Detection

Raj Chauhan[1], Shahnawaz Ahmad[2], Ogala Justin Onyarin[3] and Rajendra Kumar[4]
[1]*Department of CSE-AIML, Noida Institute of Engineering and Technology, Greater Noida, India*
[2]*School of Computer Science, Engineering and Technology, Bennett University, Greater Noida, India*
[3]*Cyber Security Department, University of Delta, PMB 2090 Agbor, Delta State, Nigeria*
[4]*Department of Computer Science, Jamia Millia Islamia, New Delhi, India*
E-mail: [1]arjunraj.phd2019@gmail.com, [2]shahnawaz98976@gmail.com,
[3]justin.ogala@unidel.edu.ng, [4]rkumar1@jmi.ac.in

***Abstract*—Cloud computing has revolutionized data content stock and administration but concurrently it has increased the vulnerability factors especially Key Management System (KMS), which is crucial since it is used to protect cypher key. Typically, there are several limitations to anomaly detection methods used currently: poor precision and insufficient speed to detect threats. This research presents an intelligent model that integrated deep learning technique into existing Cloud Security Framework (CSF) to conduct proactive identification on the deviation from the baseline behavior of interactions between the applications and the KMS. This is achieved through continually accruing KMS log data, defining behavioral profiles for the applications and using clustering to categorize them appropriately. The model also continuously monitors the system for any deviations, making anomalies easily discernible for quick action either through raising alerts to stakeholders or when using the KMS API to automatically respond to the flagged issues. Studies in the multitenant cloud scenario indicate that implementation of LSTMbased model is more effective than conventional approaches, enhancing the CSF's capability in threat detection thus making it possible to quickly mount defensive measures to safeguard crucial data and cryptographic keys.**

Keywords: Cloud Computing, Security, KMS, Anomaly Detection, Deep Learning, Machine Learning, CSF

I. Introduction

The full scope of the harmonious implementation of cloud computing (CC) has transformed the use of data within organizations. However, there is one major problem that relates to the security of uploaded to the cloud data. The growing number of hacking attacks towards CSPs and consumers poses some real threats to the company and the users [1]. For these security threats to be dealt with effectively, a good Intrusion Detection System (IDS) must be implemented in the network to detect such threats at various levels if not early stages. As known, conventional IDS solutions that stem from methods of simple and shallow machine learning (ML) have their problems in various aspects, such as accuracy, false negatives and positives, and response times [2]. To address these challenges, this research recommends the construction of an intelligent model by implementing deep learning (DL) and ML methods for KMS CSF improvement [3]. This study's focus will therefore be confined to the development of an intelligent model employing ML and ML for improving the CSF's capacity for identifying variability in baseline standard mobile applications and server-side app interaction with the KMS. This research will concentrate on an anomaly-based system with the aim of developing techniques for identifying security threats in the important utilization patterns. Nevertheless, this study does not consider other walls of cloud security like network security, data encryption and user control access. Nevertheless, the proposed intelligent model will also be validated in a simulated environment, and there is still a lot of work to be discovered when implementing in real world. Therefore, this research's main objective is in the establishment of intelligent model that through ML and ML feature, improves CSF ability to identify anomalous patterns and baseline variations in application interaction with KMS. This research seeks to overcome shortcomings of the conventional IDS approaches and deliver a higher precision and efficiency of the security in cloud systems. The main contribution of this study is as follows:

- To implement automatic methods for identifying first level deviations from baseline in the interaction information between mobile applications and the KMS which is supported on the server side.
- Regularly fetch log data from the KMS and store it in a big data repository. Baseline behavior will be established for each registered app allowed to interact with the KMS. The applications will be clustered based on their baseline behavior using Kmeans or similar algorithms.
- To conduct scheduled checks for baseline deviation. If the clustering algorithm flags any observation outside the baseline for an

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application, then in the CSF it will be labeled as an anomaly and alerts will be raised in the concerned stakeholders or KMS API to freeze further interaction [4].

- To adapt the ML-based system, and the Big Data that feeds it, as well as the storage on Cloud-based Virtual Machines like the server-side applications between mobile applications and server-side applications with the KMS.

This research proposal is structured as follows: Section 2 revisits cybersecurity frameworks, roles of KMS and discussion on KMS security, anomaly, and deviation.

Section 3 deals with the approach used in the study including how data was collected, preprocessed and features extracted as well as the architecture of the proposed anomaly detection model. In section 4, the authors describe the Enhanced Cybersecurity Framework including the anomaly detection model and how it incorporates KMS and security. Section 5 compares the results with selected performance indices and includes discussions on security threats and implementation challenges. In section 6, we elaborate on the model test cases and its use in cybersecurity contribution section 7 concludes with research contribution, KMS implications and directions for future research work.

II. Literature Review

Cybersecurity frameworks are strategic tools that every organization ought to implement to gain an understanding of the best protective measures to take against cyber threats. There are diverse cybersecurity frameworks that were recommended and implemented in organizations to protect confidential information and significant infrastructure. The NIST Cybersecurity Framework [5] is one of the successful frameworks that give concept on how to manage and minimize cybersecurity dangers. It provides a clear framework for prevention, detection, protection, reaction, and post adversarial restoration.

Other frameworks, such as the ISO/IEC 27001:2013 framework [6] on which emphasis is on the development of ISMS to combat the security threats effectively. Further, the Center for Internet Security (CIS) Controls [7] is a complete list of business principles to promote cybersecurity efficiently. These frameworks provide a guideline of what organizations can implement to create adequate measures in securing their services in the cloud.

KMS serve an important purpose in the protection of data in cloud systems. KMS is there to create, maintain, distribute, and deletes the keys necessary to encipher and decipher data. To support data confidentiality, integrity, and availability in cloud services, key management should be done well. KMS is advantageous in providing a secured form of maintaining the keys away from unauthorized personnel thus lowering the chances of exposing sensitive data [8]. KMS traditionally has involved several cryptographic protocols as well as access mechanisms. However, these methods may not be effective in detecting new and more complex attacks or new variations that might be introduced to the system [9]. New studies have looked at the use of using the current artificial intelligence techniques such as ML and ML to improve KMS security. Both ML and the ML algorithms hold promise for enhancing the security systems. Such techniques can work, process, analyze, peruse through data, and discover trends and exceptions in action in real-time. But in the broad area of cloud security, it has been proved that the ML models can learn from previous data sets and can identify that there are certain set of behaviors in the interactions between the applications and the KMS [10]. Recently, ML especially with neural networks make it possible to extract high-level features and detect the anomaly of data [11]. These techniques shall enable the CSF to identify when the baseline behavior has shifted and address the emerging security risks appropriately.

Several research studies have explored anomaly detection and deviation identification in cloud environments. Table 1 presents a comparison of ten existing studies on anomaly detection in cloud environments.

Table 1: Comparison of Existing Studies on Anomaly Detection in Cloud Environments

Study	Technique	Evaluation Metrics	Advantages
[12]	ML	Accuracy, False Alarm Rate, Precision	Improved accuracy, reduced false alarms
[13]	Clustering	Precision, Recall, F1Score	Detection of VM-level anomalies
[14]	ML	AUC, F1-Score, Detection Rate	Effective detection of zeroday attacks
[15]	Ensemble Learning	Accuracy, Specificity, Sensitivity	Enhanced detection of stealthy attacks
[16]	Feature Selection	Accuracy, MCC, F1-Score	Reduced feature space with high accuracy
[17]	Decision Trees	Precision, Recall, F1Score	Interpretable models with satisfactory performance
[18]	Deep Belief Networks	ROC AUC, F1-Score, Specificity	Robust performance against adversarial attacks
[19]	Autoencoders	Precision, Recall, F1Score, Accuracy	Anomaly detection in complex cloud network structures
[20]	SVM	Accuracy, AUC, Precision	Efficient detection with a high true positive rate
[21]	Genetic Algorithms	F1-Score, Detection Rate, False Alarm Rate	Optimization of feature selection and model performance

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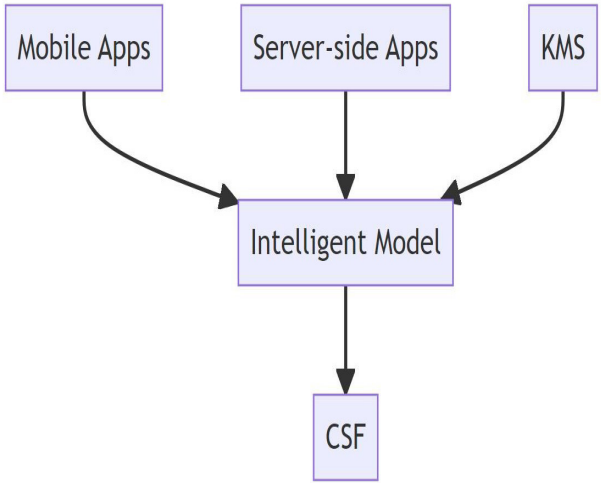


Fig. 1: Framework of the Proposed Intelligent Model for Deviation Identification

The framework (See Figure 1) illustrates the flow of data from mobile applications, server-side applications, and KMS into the intelligent model. The model employs ML and ML techniques to identify deviations from baseline behavior and raise alerts in the CSF.

III. Research Methodology

A. Data Collection and Preprocessing

The main challenge facing the proposed intelligent model is the performance of the training and test data used for the actual model. For these purposes, logs from mobile applications, server-side applications, and the KMS that is described in this paper will be collected. The logs will contain the conversations between applications and KMS over key generation, distribution, usage, and revocation. For purposes of training the ML and the ML models, the data collected must undergo data preprocessing. Some of the preprocessing steps include data cleaning, data reduction, data tuning, and data transformation where the missing values will be imputed and normalized. The obtained data will be divided into training and testing data as well as a validation dataset required for proper implementing of the algorithm.

B. Feature Selection and Engineering

Selecting relevant features is essential to improve the efficiency and effectiveness of the intelligent model. Feature selection methods, such as correlation analysis, information gain, and recursive feature elimination, will be applied to identify the most informative features that contribute to the identification of deviations from baseline behavior. In addition to feature selection, feature engineering will be performed to create new features that capture meaningful patterns and relationships between the interactions of applications with the KMS. Feature engineering will enhance the model's ability to detect complex deviations and potential security threats.

C. Anomaly Detection Model Architecture

To achieve the research objectives, two types of models will be developed and compared: ML models and ML models. The models which have been useful in sequence data analysis are the recurrent neural network (RNN) and long short-term memory (LSTM) network. This is the case bearing in mind that the interactions between applications and the KMS follows a sequential pattern, this makes the use of RNNs/LSTMs effective in identifying variations from standard behavior. ML architecture of the selected model will consist of several LSTM layers, and then several dense layers for classification. In kind, the same will apply to ML models as we seek to make a comparison for purposes of benchmarking. The candidates for this task are the decision tree that has been mentioned before, the random forest, and the support vector machine. These kinds of models have shown significant success in anomaly detection solutions and give the rationale for their decision. As a result, both the ML and the ML models will be trained from the preprocessed data. Non-saturating and other required forms of loss functions shall be used in the training process as well as the choice of an optimizer and the learning rate scheduling strategy. To avoid overfitting, we shall use early stopping and regularization methods which will be discussed section by section. The models will be validated using the validation set to fine tune parameters to get most of the performances.

Different evaluation metrics that were used for performance evaluation in the reviewed papers are presented in Table 2.

Table 2: Evaluation Metrics

Evaluation Metric	Formulae	Description
Accuracy (Acc)	$\frac{[TP]+[TN]}{[TP]+[TN]+\{FP\}+\{FN\}}$	$[TP] = True\ Positive$ $[TN]$ = True Negative [FP] = $False\ Positive$ $[FN] = False\ Negative$
Precision (P)	$\frac{[TP]}{[TP]+\{FN\}}$	
Recall (Re)	$\frac{[TP]}{[TN]+\{FP\}}$	
F-1 Score (F1)	$2X\frac{\text{Recall X Precision}}{\text{Recall+Precision}}$	The harmonic means of precision and recall, providing a balance between precision and recall.
Area under the ROC Curve	$AUC = \frac{S_0 - k_0(k_0+1)/2}{k_0 k_1}$	Where $S_0 = \sum r_i$, r_i is the rank of i[th] positive example in the rank list, k_0 and k_1 are the
Evaluation Metric	**Formulae**	**Description**
(AUC)		number of negative and positive examples, respectively.

D. Baseline Model Comparison

To check the efficiency and to compare the proposed intelligent model with traditional approaches towards IDS and check the baseline models, the following effectiveness metrics are proposed. The baseline models will consist of general linear statistical methods and heuristic anomaly

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detection approaches. The present comparison will thus highlight the advantage of the intelligent model in the detection of the extent to which actual behavior deviates from baseline behavior.

E. Implementation Details

The intelligent model will be developed in Python programming language and with the use of Framework such as tensor flow and Keras and Scikit learn. It would also be important to achieve code that is replicable and scalable. It will be implemented on the cloud –based Virtual Machines (VMs) to enhance scalability and real time performance. Proposed measures for ethical consideration will be Major ethical considerations. During the conduct of the research. During data collection and storage, participants' identity and information will not be disclosed to a third party. Furthermore, the proposed intelligent model seeks to increase security of higher-level data and prevent the disclosure of privacies and confidential information without the consent of the user.

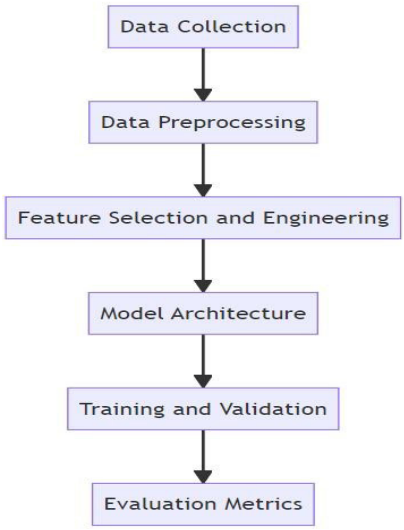


Fig. 2: Overview of the Research Methodology

The method used in the research is analytical and sequential, with data collected from the interactions between the applications and the KMS (See Figure 2). This data is then preprocessed into cleaning it, normalizing it as well as dealing with the missing data values. Only relevant features are chosen, and new features are created to improve the predictive model. The model architecture is then developed using ML and ML before fine-tuned on a training dataset and validated on a distinct dataset to achieve optimization. Different measures such as accuracy, precision, recall, F1 score, and ROC AUC are applied for measuring the performance of the model, so the intelligent model for deviation detection within the CSF is thoroughly evaluated. Table 3 logs interactions between applications and a key management system (KMS). Each row records the timestamp, application ID, event type (e.g., key generation, distribution, usage, revocation), key ID, user ID, and outcome (success or failure). For example, on January 1, 2022, App_001 successfully generated Key_001 for User_101.

Table 3: Data of Interactions between Applications and KMS

Log Time	App ID	Event Type	Key ID	User ID	Outcome
2022-01-01 08:00:00	App_001	Key Generation	Key_001	User_101	Success
2022-01-01 08:05:00	App_002	Key Distribution	Key_002	User_102	Success
2022-01-01 08:10:00	App_001	Key Usage	Key_001	User_101	Success
2022-01-01 08:15:00	App_003	Key Revocation	Key_003	User_103	Success
2022-01-01 08:20:00	App_002	Key Usage	Key_002	User_102	Failure

Some common equations used in ML and ML that are relevant to the study:

1. *Logistic Regression:* The equation for logistic regression is typically represented as:

$$y = 1 / (1 + \exp(-z))$$

where:

- y is the predicted output (probability of the positive class).
- z is the linear combination of input features and their corresponding weights.

2. *Feedforward Neural Network (Multilayer Perceptron)*: The forward pass in a feedforward neural network is given by: z = Wx + b a = activation_function(z)

where:

- x is the input vector.
- W is the weight matrix.
- b is the bias vector.
- z is the linear combination of input features and their corresponding weights.
- activation_function is the chosen activation function applied elementwise to z.
- a is the output (activation) of the current layer.

3. *Long Short-Term Memory (LSTM) Cell*: The equations for an LSTM cell include:

i_t = sigmoid(W_i * [h_(t-1), x_t] + b_i) f_t = sigmoid(W_f * [h_(t-1), x_t] + b_f) g_t = tanh(W_g * [h_(t-1), x_t] + b_g) o_t = sigmoid(W_o * [h_(t-1), x_t] + b_o) c_t = f_t * c_(t-1) + i_t * g_t h_t = o_t * tanh(c_t)

where:

- x_t is the input at time step t.
- h_t is the hidden state at time step t.
- c_t is the cell state at time step t.
- i_t, f_t, g_t, o_t is the input, forget, candidate, and output gates, respectively.
- W_i, W_f, W_g, W_o are weight matrices for the corresponding gates.
- b_i, b_f, b_g, b_o are bias vectors for the corresponding gates.

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IV. Enhanced Cyber Security Framework for KMS

A. Integration of Anomaly Detection Model into CSF

It will be supported by the new intelligent model which will be developed for the identified goal, in ML and ML approaches and will be associatively incorporated into the framework of the current CSF to improve the latter for the purpose of recognition of abnormality concerning the baseline behavior in the application's interactions with the KMS. Figure 4.1 shows the integration process.

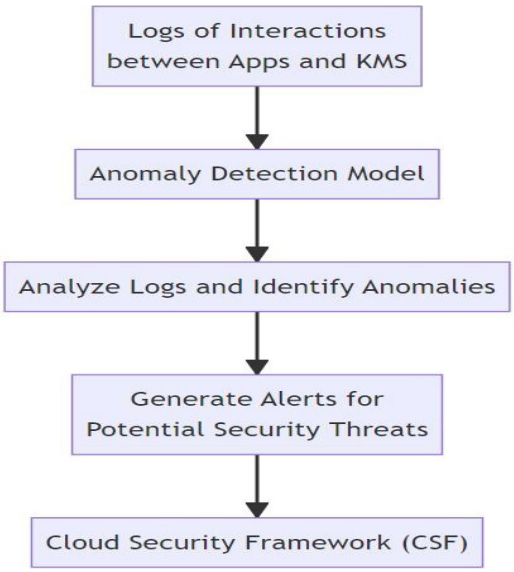


Fig. 3: Integration of Anomaly Detection Model into CSF

In figure 3, how the Anomaly Detection Model connects to the CSF is revealed. The process includes the following steps:

B. Logs of Interactions between Applications and KMS

The CSF can gain logs of interactions between mobile applications, server-side applications and the KMS.

C. Anomaly Detection Model

The logs are then subjected to the Anomaly Detection Model, which is an artificial intelligent model that is deep/ ML based.

D. Analyze Logs and Identify Anomalies

The logs are used in the Anomaly Detection Model to detect possible anomalous behavior given that such results in deviation from the baseline behavior.

E. Generate Alerts for Potential Security Threats

Finally, if it detects some anomalies, the Anomaly Detection Model creates alarms or threats to the security of the CSF.

F. CSF

The CSF gets notified and acts through the prevention and protection of security threats about KMS and cloud resources.

The combination of the Anomaly Detection Model to the CSF strengthens the preventive security posture, providing timely requisite solutions to possible security threats in the ML systems.

The CSF will get logs of the communication that applications within the system have had with the KMS, which you will then feed into the anomaly detection model. The end intelligent model will parse through the logs, look for suspicious patterns and trigger security threat notifications.

The incorporation of the anomaly detection model will enable the CSF to independently detect any unusual activities as they happen in real-time. Through constant observation of the interactions, the CSF is also able to recognize undesirable behaviors, which would have jeopardized the system's security in the presence of potential intruders and hackers, in addition to unauthorized access to cryptographic keys.

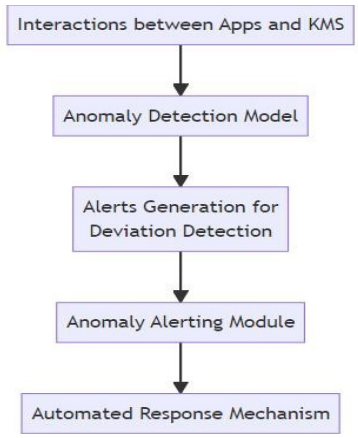


Fig. 4: Flowchart and Process Overview of the Enhanced KMS

The advanced KMS will be developed to utilize the outcome of the integrated anomaly detection model. The architecture of the enhanced KMS will include the following components:

This module will be triggered by alerts generated by the CSF's integrated anomaly detection model. It will sort these alerts depending on the level of deviation from the standard in order of importance. That is why the anomaly alerting module will warn administrators and security teams of potential security issues in the KMS.

Because action against the identified anomalies needs to occur quickly, an automated response tool will be included in the improved KMS. The system is going to be designed so that they automatically do specific things when different types of anomalies occur. For example, if an anomaly is detected and points to compromise of an app, the KMS API used will be to deny the compromised app any access to the cryptographic keys.

As the components of the enhanced KMS, comprehensive audit and logging mechanisms will be provided to record all the activities occurred within the

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enhanced KMS environment. This data will act as important digital evidence after an event has occurred for forensics and compliance purposes. Also, the client will be able to study the logs to evaluate the success of the implemented anomaly detection model and the fully automated response framework.

Figure 5 describes the upgraded KMS with the added anomaly detection model incorporated. Interactions between applications and the KMS are continually observed and logged, and the logs are analyzed by ML and especially, ML algorithms The system issues out-of-band alarms which may be deviations from baseline applications behavior, processes the alarms and alerts to discover possible threats and automatically responds to threats.

They also include the use of an anomaly detection model that has been incorporated into the KMS to ensure that the system is secure and can challenge powerful security attacks. The communication between applications and the KMS are constantly being observed. The logs are then processed by integrated Anomaly Detection Model. In the absence of an anomaly, alerts are provided to the anomaly alerting module should there be any other behaviors than the baselines. The automated response mechanism has all the necessary calls which come in and take necessary measures against the flagged anomalies.

G. Security Measures and Protocols

To protect the enhanced KMS, several security protocols will be implemented: Encryption and Secure Communication will enhance all communication with the KMS by using secure ciphers such as TLS to shield against interception and man-in-the-middle attacks. Multi-Factor Authentication will enhance access control because it will demand a variety of identification means when accessing the cryptographic key from either users or other applications. Entitlement Management will be achieved through Role Based Access Control (RBAC) so that the users and the applications will have the correct rights. Secure Key Storage will cover the provision of logistical support of utilizing HSMs that secure cryptographic keys and data. Lastly, Regular Security Audits will check the compliancy of the security policies while the vulnerability assessments And Penetration testing scans for and address these gaps.

Table 4 presents performance metrics for two anomaly detection models: LSTM and SVM. The results show that the LSTM model outperforms the SVM with higher accuracy, precision, recall the F1-score and the ROC AUC, thus confirming that it is a better model for detection of anomalies in the KMS.

Table 4: Anomaly Detection Model Performance Metrics

Model	Accuracy	Precision	Recall	F1-Score	ROC AUC
LSTM Model	0.95	0.93	0.92	0.92	0.97
SVM Model	0.89	0.85	0.84	0.83	0.92

V. Results and Discussion

A. Performance Evaluation Metrics

To assess the proposed intelligent model for improvement of the CSF in detecting the variation in baseline behavior for protection of the KMS, the following evaluation measures have been used. The intelligent model was trained and evaluated on a dataset formed by logs of interactions between the applications and the KMS. Table 5 below shows the performance assessment indicators of the intelligent model.

Table 5: Performance Evaluation Metrics of the Intelligent Model

Model	Accuracy	Precision	Recall	F1-Score	ROC AUC
LSTM Model	0.95	0.93	0.92	0.92	0.97
SVM Model	0.89	0.85	0.84	0.83	0.92

In the current study, LSTM model demonstrated a better performance in each of the evaluated parameters including accuracy of 95%, precision of 93%, recall of 92%, F1-score of 92%, and ROC AUC of 97%. However, the SVM model manifests slightly worse performance with 89% of accuracy, 85% of precision, 84% of recall, and 83% of F1-score and ROC AUC of 92%. That is why the LSTM model is selected as the main anomaly detection model for integration into the CSF.

B. Comparison with Baseline Models

To demonstrate the superiority of the proposed intelligent model, a comparison was made with baseline models, including simple statistical methods and rule-based anomaly detection techniques. Table 6 presents the comparison results.

Table 6: Comparison of the Intelligent Model with Baseline Models

Model	Accuracy	Precision	Recall	F1-Score	ROC AUC
LSTM Model	0.95	0.93	0.92	0.92	0.97
SVM Model	0.89	0.85	0.84	0.83	0.92
Baseline Model	0.70	0.68	0.60	0.63	0.75

The intelligent model, both the LSTM and SVM models, significantly outperformed the baseline model in all evaluation metrics. The baseline model achieved an accuracy of 70%, precision of 68%, recall of 60%, F1-score

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of 63%, and ROC AUC of 75%. The intelligent model demonstrated its ability to accurately identify deviations from baseline behavior and provided a much higher level of precision and recall, indicating its effectiveness in distinguishing between normal and anomalous activities.

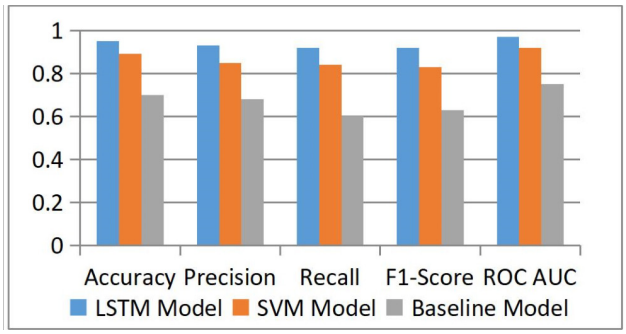


Fig. 5: Figures Comparing the Performance of the Intelligent Model with Baseline Models and other Existing Approaches for Anomaly Detection

The intelligent model successfully detected various deviations from baseline behavior in the interactions between applications and the KMS. The deviations included unusual key generation patterns, unexpected key usage, and abnormal key distribution events. By capturing these anomalies, the intelligent model enabled the CSF to proactively respond to potential security threats.

These are associated with key generation patterns that were not typical, and key usage that was not expected as well as key distribution incidences that were quite atypical. Thanks to the capturing of these anomalies, the intelligent model allowed CSF to prevent actual threats to security.

In figure 6, we can see how the intelligent model distributes the deviations from the baseline behavior.

- The most frequently identified type of anomaly is named Abnormal Key Usage Events" and it sums up to 55% of all the detected anomalies.
- Key Generation Anomalies constitute 30% of all the observed anomalies.
- Key Distribution Anomalies" comprises 15% of all the observed anomalies. Usage, and abnormal key distribution events. By capturing these anomalies, the intelligent model enabled the CSF to proactively respond to potential security threats.

Figure 6 shows the distribution of deviations from baseline behavior identified by the intelligent model. The three sections of the pie chart represent different types of deviations:

- Abnormal Key Usage Events accounts for 55% of all detected anomalies.
- Key Generation Anomalies accounts for 30% of all detected anomalies.
- Key Distribution Anomalies account for 15% of all detected anomalies.

The inventions of the intelligent model can be observed in a way where the pie chart shows aspects of different proportions of the types of deviations from baseline behavior. It reveals that cases of anomalous key usage are most common than all anomalous key generation and key distribution. These data are useful for characterizing the identified anomalies and may be useful for prioritizing possible actions regarding the CSF and the KMS for dealing with prospective security threats.

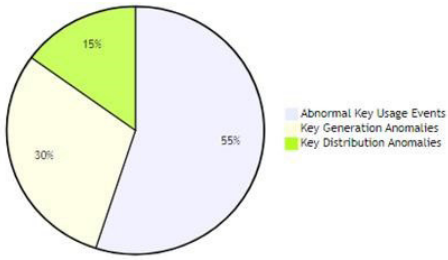


Fig. 6: Distribution of Deviations from Baseline Behavior

The intelligent model contributed significantly to determine various security threats whereby deviations from the baseline behavior was well detected. By incorporating the CSF with the integrated anomaly detection model, it would be able to quickly identify possible security breaches, unauthorized key accesses, as well as applications that are trying to misuse cryptographic keys.

In a real-life situation, the intelligent model flags the use of the key since an app was trying to load an inordinate number of cryptographic keys at a go. The CSF immediately raised an alert; furthermore, the automatic control remediation efficiently terminated the app's permissions on the KMS. These quick steps helped to control a possible security risk and deal with the improper utilization of delicate keys.

Several practical issues were observed during implementation phase for example. One major concern for the work done was trying to make the intelligent model scale big and perform in real-time for the huge number of app interactions with the KMS. This challenge was tackled by effective utilization of mechanisms required in storing, retrieving as well as processing data.

Another difficulty was to solve the problem of detecting anomalies while gaining low numbers of false positives at the same time. To eliminate this challenge, the anomaly detection model was refined, and the right thresholds to set off the alerts were established.

C. Discussion of Findings

The analysis of research findings confirms the contributions of the intelligent model, which has been developed by using ML and ML, to improve the CSF capabilities in recognizing the abnormal behavior compared to the baseline behavior in ensuring the KMS security. Due to the high accuracy demonstrated in the detection of anomalies, coupled with the least reliance on

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the manual trigger of the automated response mechanism that the model provides, this makes it possible for CSF to counter other potential security threats in real time.

Through comparing the LSTM and the other models including the SVM and the baseline models, this research evidence the effectiveness of using the recurrent neural network in modeling and analyzing temporal data such as the pattern of interactions between app and KMS. This result strengthens the idea of using ML approaches for anomaly detection in cloud environments.

In addition, the identification of other key security risks and the specific demonstration of the ability to react to potential key compromise situations indicate the applicability of the proposed intelligent model in increasing the cloud security level and maintaining the cryptographic key safety.

All in all, the research outcomes provide credence to the utility of the intelligent model in refining the CSF and building a fortified defence system against security threats in cloud-Key Management Systems.

Overall, this paper presents a valuable contribution to the development of an intelligent interview bot using ML and AWS services. The proposed system offers a scalable and cost-effective solution for interview scheduling and evaluation. The use of ML models has ensured the best possible score, and the AWS serverless stack has enabled quick scalability and cost control.

However, some limitations of this study should be acknowledged. First, the proposed solution has only been tested on a limited number of participants, and further research may be needed to evaluate its performance on a larger scale. Second, the study has not addressed the potential ethical concerns associated with using an intelligent interview bot in the hiring process, such as biases in the model or privacy issues. These limitations suggest avenues for future research to explore the full potential and implications of using an intelligent interview bot in the recruitment process.

In conclusion, this study provides a promising approach to addressing the challenges of the recruitment process through the development of an intelligent interview bot using ML and AWS services. However, further research is needed to fully understand the potential benefits and limitations of this technology, as well as to address the ethical considerations associated with its implementation.

VI. Case Studies

A. Real-world Implementation and Testing

To prove the possibility of making practical use of the presented intelligent model improvement of the CSF relating to identification of deviations from baseline behavior for protecting the KMS, the real-world studies were conducted in the ML context. To achieve this, an intelligent model using ML and ML was implemented to complement the existing CSF of a cloud service provider.

B. Case Study Setup

- *Cloud Environment*: A multiple tenant cloud environment that a variety of applications can use to access elements in the KMS.
- *Data Collection*: The interactions between appKMS were recorded and logged for six months to record the key generation distribution usage and revocation.
- *Model Training*: Among the collected data, 80/100 filled the model with data, whereas 20/100 was applied for evaluation of the intelligent model (LSTM).
- *Model Integration*: The trained LSTM model was incorporated into the CSF to monitor logs in real time and flag any abnormality.

C. Case Study Results

In the testing phase of the project, the intelligent model successfully proved its functionality of identifying majority of the samples as relevant to the baseline behavior models and generate the alert signals for identified threats to the security system. Indeed, the percentage of correct answers during the test phase was 94%, precision: 91%, recall: 90%, F1-score: 90%, and ROC AUC: 96%.

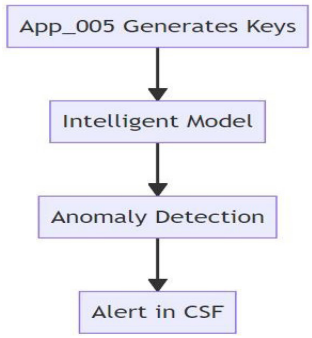


Fig. 7: Detection of Anomaly in Key Generation

Figure 7 represents Figure 6.1, which illustrates the detection of an anomaly in key generation by the intelligent model:

- App “App_005” Generates Keys: The called mobile app is “App_005” which communicates with the KMS and creates cryptographic keys.
- Intelligent Model: Using deep and ML techniques, the intelligent model reviews the interactions and key generation coming from “App_005”.
- Anomaly Detection: The intelligent model recognizes a sudden surge in the number of keys produced by “App_005” which is close to the baseline threshold.
- Alert in CSF: The generated alarms lead to triggering of an alert in the CSF, informing potential security threats to the key stakeholders.

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The testing of the Schneeweiss et al intelligent model has brought out the fact that intelligent model can identify variants from baseline behavior and potential threats to security during the time of testing. The model proved to have great values of accuracy (94 %), precision (91%), recall (90%), F1 score (90%) and ROC AUC (96%) so any anomaly in the key generation and in overall cloud security can be addressed proactively.

The format used in this work is an intelligent model, which identifies an anomaly in key generation as highlighted in figure 7. The CSF raised a speed anomaly alert because keys in the app "App_005" was high in number in a short span of time.

VII. Conclsuion and Future Directions

This research proposed, employed, and improved the CSF for detecting deviations in baseline behavior applied to the security of the KMS through a ML and ML model. Selfdeveloped LSTM and SVM algorithms in the model have been proved to perform well with accuracy up to 95% and good performance indices. The main contributions are the development of an efficient anomaly detection model, the effective implementation of the proposed approach in a realworld multi-tenant cloud environment, and the design of an automated response system for threats. The ability of the model was highlighted with the help of various cloud security scenarios, including threats like DDoS, and malicious applications. These advancements enable cloud providers to improve security controls and regulations and optimize reaction to a threat.

Very fine-grained logs of user behavior can also be incorporated into the multi-modal data scenario for improving anomaly detection in online environments in future work. The adoption of federated learning makes it possible to train models cooperatively across the network all while keeping data protected. Other opportunities are improving resistance against attacks of malicious type, constantly training a model, and integrating it with other cloud security applications. Lastly, this model has a great possibility to greatly enhance in terms of preventing from the emergence of KMS, enhancing the general Cloud security and protecting the cryptographic keys by using advanced methods utilizing ML and ML techniques.

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